

# Homework week 7, core econometrics, fall 2025

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## 1. Partial identification:

Suppose the following assumptions hold:

- (a)  $Y = D \cdot Y^1 + (1 - D) \cdot Y^0$ , where  $D \in \{0, 1\}$  and  $Y \in [0, 1]$ ,
- (b)  $D = Z \cdot D^1 + (1 - Z) \cdot D^0$ , where  $Z \in \{0, 1\}$ ,
- (c)  $D^1 \geq D^0$ ,
- (d)  $Z \perp (Y^0, Y^1, D^0, D^1)$ .

Provide explicit bounds on the CDF of potential outcomes,

$$P(Y^d \leq y),$$

for  $d = 0, 1$ . Discuss these bounds in terms of “never takers”, “always takers”, and “compliers”.

## 2. Inverse probability weighting:

Suppose the following assumptions hold:

- (a)  $D \in \{0, 1\}$
- (b)  $Y = D \cdot Y^1 + (1 - D) \cdot Y^0$
- (c)  $(Y^1, Y^0) \perp D | X$
- (d)  $p(X) < 1$  for almost all  $X$ , where  $p(X) = P(D = 1 | X)$

Find a weight function  $w(D, X)$ , such that

$$E[w(D, X) \cdot \mathbf{1}(Y \leq y)] = P(Y^0 \leq y | D = 1).$$

## 3. Failure of standard asymptotics:

Consider i.i.d. draws of a random variable  $X_i$  with mean  $\mu$  and variance  $\sigma^2$ .

- (a) What is the asymptotic distribution of  $\sqrt{n}(\bar{X}_n - \mu)$ , where  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ ?

- (b) What is the asymptotic distribution of  $\sqrt{n}(|\bar{X}_n| - |\mu|)$ , for  $\mu \neq 0$ ?  
(Hint: Use the delta method.)
- (c) What happens when  $\mu = 0$ ?  
What happens when  $\mu$  is close to 0?  
(Hint: Explicitly consider the remainder of the first-order approximation used for the delta method.)
- (d) Discuss the bias and deviation from normality of the distribution of  $\bar{X}_n$ , assuming that  $X_i \sim N(\mu, \sigma^2)$ , so that the approximation in item (a) is exact.